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Inventor(s)

SUN; Yangyang et al.

### PACKAGE COMPRISING A BASE SUBSTRATE AND INTEGRATED DEVICES

#### Abstract

A package comprising a base substrate; a first integrated device coupled to the base substrate through a bonding material; a second integrated device coupled to the base substrate through the bonding material; a fill material coupled to the base substrate, the first integrated device and the second integrated device; and a metallization portion coupled to the first integrated device and the second integrated device.

**Inventors:** SUN; Yangyang (San Diego, CA), HE; Dongming (San Diego, CA), LI; Yue (San Diego, CA)

**Applicant:** QUALCOMM Incorporated (San Diego, CA)

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## Background/Summary

### FIELD

[0001] Various features relate to devices with integrated devices.

### BACKGROUND

[0002] A package may include a substrate and integrated devices. These components are coupled together to provide a package that may perform various electrical functions. There is an ongoing need to provide better performing packages. There is also an ongoing need to reduce the overall size of the packages.

### SUMMARY

[0003] Various features relate to devices with integrated devices.

[0004] One example provides a package comprising a base substrate; a first integrated device coupled to the base substrate through a bonding material; a second integrated device coupled to the base substrate through the bonding material; a fill material coupled to the base substrate, the first integrated device and the second integrated device; and a metallization portion coupled to the first integrated device and the second integrated device.

[0005] Another example provides a method for fabricating a package. The method provides a base substrate. The method couples a first integrated device to the base substrate through a bonding material. The method couples a second integrated device to the base substrate through the bonding material. The method couples a fill material to the base substrate, the first integrated device and the second integrated device. The method couples a metallization portion to the first integrated device and the second integrated device.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Various features, nature and advantages may become apparent from the detailed description set forth below when taken in conjunction with the drawings in which like reference characters identify correspondingly throughout.

[0007] FIG. 1 illustrates an exemplary cross sectional profile view of a package that includes a base substrate, integrated devices and a metallization portion.

[0008] FIG. 2 illustrates an exemplary cross sectional profile view of a package that includes a base substrate, integrated devices and a metallization portion.

[0009] FIG. 3 illustrates an exemplary cross sectional profile view of a package that includes a base substrate, integrated devices and a metallization portion.

[0010] FIG. 4 illustrates an exemplary cross sectional profile view of an integrated device.

[0011] FIGS. 5A-5C illustrate an exemplary sequence for fabricating a package that includes a base substrate, integrated devices and a metallization portion.

[0012] FIG. 6 illustrates an exemplary flow chart of a method for fabricating a package that includes a base substrate, integrated devices and a metallization portion.

[0013] FIGS. 7A-7C illustrate an exemplary sequence for fabricating a metallization portion.

[0014] FIG. 8 illustrates an exemplary cross sectional profile view of a redistribution portion.

[0015] FIG. 9 illustrates an exemplary flow chart of a method for fabricating a metallization portion.

[0016] FIGS. 10A-10B illustrate an exemplary sequence for fabricating a metallization portion.

[0017] FIG. 11 illustrates various electronic devices that may integrate a die, an electronic circuit, an integrated device, an integrated passive device (IPD), a passive component, a package, and/or a device package described herein.

#### DETAILED DESCRIPTION

[0018] In the following description, specific details are given to provide a thorough understanding of the various aspects of the disclosure. However, it will be understood by one of ordinary skill in the art that the aspects may be practiced without these specific details. For example, circuits may be shown in block diagrams in order to avoid obscuring the aspects in unnecessary detail. In other instances, well-known circuits, structures and techniques may not be shown in detail in order not to obscure the aspects of the disclosure.

[0019] The present disclosure describes a package comprising a base substrate; a first integrated device coupled to the base substrate through a bonding material; a second integrated device coupled to the base substrate through the bonding material; a fill material coupled to the base substrate, the first integrated device and the second integrated device; and a metallization portion coupled to the first integrated device and the second integrated device. The bonding material and/or the fill material may include one or more inorganic materials. As will be further described below, the package has a compact form factor while providing high density interconnections, improved thermal performance and/or improved performance.

Exemplary Package Comprising a Base Substrate, Integrated Devices and a Metallization Portion

[0020] FIG. 1 illustrates a cross sectional profile view of a package **100** that includes a base substrate **102**, an integrated device **103**, an integrated device **105**, a fill material **106**, a metallization portion **104**, a plurality of pillar interconnects **107**, a plurality of solder interconnects **109**, a bonding material **135** and a bonding material **155**.

[0021] The base substrate **102** may include silicon (Si) (e.g., silicon wafer). The integrated device **103** includes a front side and a back side. The integrated device **103** comprises a plurality of pads **130**. The front side of the integrated device **103** may include the side that comprises the plurality of pads **130**. The back side of the integrated device **103** may include the side that comprises a die substrate. The integrated device **105** includes a front side and a back side. The integrated device **105** comprises a plurality of pads **150**. The front side of the integrated device **105** may include the side that comprises the plurality of pads **150**. The back side of the integrated device **105** may include the side that comprises a die substrate. A more detailed example of an integrated device is illustrated and described below in at least FIG. 4.

[0022] The integrated device **103** is coupled to the base substrate **102**. The integrated device **103** may be coupled to the base substrate **102** through the bonding material **135**. The back side of the integrated device **103** may be coupled to the base substrate **102** through the bonding material **135**. The bonding material **135** may include oxide and/or silicon. The bonding material **135** may be coupled to and touch (i) the base substrate **102** and (ii) the back side of the integrated device **103**. For example the bonding material **135** may be coupled to and touch the die substrate of the integrated device **103**. The integrated device **103** may be a first integrated device. The bonding material **135** may be a first bonding material. The bonding material **135** may include an inorganic material. In some implementations, the thickness of the integrated device **103** may have a thickness in a range of about 10-50 micrometers (e.g., 10-20 micrometers).

[0023] The integrated device **105** is coupled to the base substrate **102**. The integrated device **105** may be coupled to the base substrate **102** through the bonding material **155**. The back side of the integrated device **105** may be coupled to the base substrate **102** through the bonding material **155**. The bonding material **155** may include oxide and/or silicon. The bonding material **155** may be coupled to and touch (i) the base substrate **102** and (ii) the back side of the integrated device **105**. For example the bonding material **155** may be coupled to and touch the die substrate of the integrated device **105**. The integrated device **105** may be a second integrated device. The bonding material **155** may be a second bonding material. The bonding material **155** may include an

inorganic material. In some implementations, the thickness of the integrated device **105** may have a thickness in a range of about 10-50 micrometers (e.g., 10-20 micrometers).

[0024] The metallization portion **104** includes a plurality of metallization interconnects **142** and at least one dielectric layer **140**. The metallization portion **104** is coupled to the front side of the integrated device **103** and the front side of the integrated device **105**. The plurality of metallization interconnects **142** may be coupled to and touch the plurality of pads **130** of the integrated device **103** and the plurality of pads **150** of the integrated device **105**. The metallization portion **104** may include a redistribution portion that includes a plurality of redistribution interconnects. The plurality of metallization interconnects **142** may include a plurality of redistribution interconnects. The plurality of metallization interconnects **142** may include a plurality of damascene interconnects. In some implementations, the plurality of metallization interconnects **142** may comprise interconnects with width and/or spacing that are less than 1 micron. In some implementations, the plurality of metallization interconnects **142** may have a minimum pitch of about 25 nanometers (nm). Different processes may be used to form the metallization portion **104**. Examples of processes for forming the metallization portion **104** are further described below in at least FIGS. 7A-7C and/or FIGS. 10A-10B.

[0025] The fill material **106** is located between the base substrate **102** and the metallization portion **104**. The fill material **106** may be coupled to and touch the base substrate **102** and the metallization portion **104**. The fill material **106** may at least partially encapsulate the integrated device **103** and the integrated device **105**. The fill material **106** may at least laterally surround the integrated device **103**. The fill material **106** may at least laterally surround the integrated device **105**. The fill material **106** may touch a side surface of the integrated device **103** and/or a side surface of the integrated device **105**. The fill material **106** may include an inorganic material. The fill material **106** may include silicon oxide. In some implementations, as used in the disclosure, silicon oxide may refer to SiO<sub>x</sub>, where x may be one or greater. For example, silicon oxide may also mean to include silicon dioxide. The fill material **106** may include the same material and/or similar materials as the bonding material **135** and/or the bonding material **155**. The use of the fill material **106**, the bonding material **135** and/or the bonding material **155** comprising an inorganic material, which has better thermal properties than an organic material, may help improve the performance of the integrated devices and/or the package by providing improved heat dissipation for the integrated devices and/or the package.

[0026] The fill material **106** may include an encapsulation layer, such as a mold, a resin and/or an epoxy. The fill material **106** may be a means for encapsulation. The fill material **106** may be provided by using a compression and transfer molding process, a sheet molding process, or a liquid molding process. An organic material may include polyimide, an underfill and/or a molding compound.

[0027] The plurality of pillar interconnects **107** are coupled to the metallization portion **104**. The plurality of pillar interconnects **107** may be coupled to the plurality of metallization interconnects **142**. The plurality of solder interconnects **109** are coupled to the plurality of pillar interconnects **107**. The package **100** may be an integrated device package. In some implementations, the package **100** may be coupled to a board (e.g., printed circuit board) through the plurality of pillar interconnects **107** and the plurality of solder interconnects **109**.

[0028] An electrical path between the integrated device **103** and the integrated device **105** may include a pad from the plurality of pads **130**, at least one metallization interconnect from the plurality of metallization interconnects **142** and/or a pad from the plurality of pads **150**.

[0029] An electrical path between the integrated device **103** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **130**, at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0030] An electrical path between the integrated device **105** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **150**, at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0031] Power to the integrated device **103** may be provided through the front side of the integrated device **103**. Power to the integrated device **105** may be provided through the front side of the integrated device **105**.

[0032] It is noted that in some implementations, the base substrate **102**, the bonding material **135** and/or the bonding material **155** may be optional. Different implementations may have different thicknesses for the base substrate **102**.

[0033] The package **100** provides several technical advantages. One, the package **100** provides improved thermal performance of the package, which helps improve the overall performance of the package **100**. Two, the metallization portion **104** helps provide fine pitch integrated device to integrated device electrical connections, which can help improve the performance of the package **100**. Three, providing power to the integrated device through the front side of the integrated device helps provide improved power delivery power, which may improve the performance of the package **100**. Four, the structure and/or the configuration of the package may be fabricated using a more integrated fabrication process, which can help lower the cost of the package and/or improve the fabrication yield of the package. Five, the use of inorganic materials for the bonding material **135**, the bonding material **155** and/or the fill material **106**, helps provide improved thermal performance (e.g., improved heat dissipation), which may help improve the performance of the package.

[0034] FIG. **2** illustrates a cross sectional profile view of a package **200** that includes a base substrate **102**, an integrated device **103**, an integrated device **105**, a fill material **106**, a metallization portion **204**, a plurality of pillar interconnects **107**, a plurality of solder interconnects **109**, a bonding material **135** and a bonding material **155**.

[0035] The package **200** of FIG. **2** is similar to the package **100** of FIG. **1**, and may include components that are similar and/or arranged in a similar manner, as described for the package **100**. The package **200** includes the metallization portion **204**. The metallization portion **204** is similar to the metallization portion **104**. However, the metallization portion **204** includes additional passive components and/or interconnects configured as passive components. The metallization portion **204** includes at least one dielectric layer **140**, a plurality of metallization interconnects **142**, a passive device **247** and a passive device **248**. The plurality of metallization interconnects **142** include at least one metallization interconnect **245** and at least one metallization interconnect **246**.

[0036] The passive device **247** may include a capacitor. The capacitor may include a metal-insulator-metal (MIM) capacitor. The passive device **248** may include a capacitor. The capacitor may include a metal-insulator-metal (MIM) capacitor. The at least one metallization interconnect **245** may be configured as an inductor. The at least one metallization interconnect **246** may be configured as an inductor.

[0037] The passive device **247** may be coupled to the integrated device **103**. For example, the passive device **247** (e.g., capacitor) may be coupled to one or more pads from the plurality of pads **130** of the integrated device **103**. The capacitor **248** may be coupled to the integrated device **105**. For example, the capacitor **248** (e.g., capacitor) may be coupled to one or more pads from the plurality of pads **150** of the integrated device **105**. The passive device **247** and/or the capacitor **248** may be considered part of the metallization portion **204**. The passive device **247** and/or the capacitor **248** may be located in the metallization portion **204**.

[0038] The metallization portion **204** is coupled to the front side of the integrated device **103** and the front side of the integrated device **105**. The plurality of metallization interconnects **142** may be coupled to and touch the plurality of pads **130** of the integrated device **103** and the plurality of pads **150** of the integrated device **105**. The metallization portion **204** may include a redistribution

portion that includes a plurality of redistribution interconnects. The plurality of metallization interconnects **142** may include a plurality of redistribution interconnects. The plurality of metallization interconnects **142** may include a plurality of damascene interconnects. In some implementations, the plurality of metallization interconnects comprises interconnects with width and/or spacing that are less than 1 micron. In some implementations, the plurality of metallization interconnects comprises interconnects with minimum width and/or spacing of about 25 nanometers (nm). In some implementations, the plurality of metallization interconnects comprises interconnects with a minimum pitch of about 25 nanometers (nm).

[0039] The fill material **106** is located between the base substrate **102** and the metallization portion **204**. The fill material **106** may touch the base substrate **102** and the metallization portion **204**. The fill material **106** may at least partially encapsulate the integrated device **103** and the integrated device **105**. The fill material **106** may at least laterally surround the integrated device **103**. The fill material **106** may at least laterally surround the integrated device **105**. The fill material **106** may touch a side surface of the integrated device **103** and/or a side surface of the integrated device **105**. The fill material **106** include inorganic material. The fill material **106** may include silicon oxide. The fill material **106** may include an encapsulation layer, such as a mold, a resin and/or an epoxy. The fill material **106** may be a means for encapsulation. The fill material **106** may be provided by using a compression and transfer molding process, a sheet molding process, or a liquid molding process.

[0040] The integrated device **103** is coupled to the base substrate **102**. The integrated device **103** may be coupled to the base substrate **102** through the bonding material **135**. The back side of the integrated device **103** may be coupled to the base substrate **102** through the bonding material **135**. The bonding material **135** may include oxide and/or silicon. The bonding material **135** may be coupled to and touch (i) the base substrate and (ii) the back side of the integrated device **103**. For example the bonding material **135** may be coupled to and touch the die substrate of the integrated device **103**. It is noted that in some implementations, the base substrate **102**, the bonding material **135** and/or the bonding material **155** may be optional.

[0041] The integrated device **105** is coupled to the base substrate **102**. The integrated device **105** may be coupled to the base substrate **102** through the bonding material **155**. The back side of the integrated device **105** may be coupled to the base substrate **102** through the bonding material **155**. The bonding material **155** may include oxide and/or silicon. The bonding material **155** may be coupled to and touch (i) the base substrate and (ii) the back side of the integrated device **105**. For example the bonding material **155** may be coupled to and touch the die substrate of the integrated device **105**.

[0042] An electrical path between the integrated device **103** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **130**, the passive device **247**, at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0043] An electrical path between the integrated device **103** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **130**, the at least one metallization interconnect **245** (e.g., configured as an inductor), at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0044] An electrical path between the integrated device **103** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **130**, the at least one metallization interconnect **245** (e.g., configured as an inductor), the passive device **247**, at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0045] An electrical path between the integrated device **105** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **150**, the passive device **248**, at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0046] An electrical path between the integrated device **105** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **150**, the at least one metallization interconnect **246** (e.g., configured as an inductor), at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0047] An electrical path between the integrated device **105** and the plurality of solder interconnects **109** may include a pad from the plurality of pads **150**, the at least one metallization interconnect **246** (e.g., configured as an inductor), the passive device **248**, at least one metallization interconnects from the plurality of metallization interconnects **142**, a pillar interconnect from the plurality of pillar interconnects **107** and/or a solder interconnect from the plurality of solder interconnects **109**.

[0048] FIG. **3** illustrates a cross sectional profile view of a package **300** that includes a base substrate **302**, an integrated device **103**, an integrated device **105**, a fill material **106**, a metallization portion **204**, a plurality of pillar interconnects **107**, a plurality of solder interconnects **109**, a bonding material **135** and a bonding material **155**.

[0049] The package **300** of FIG. **3** is similar to the package **100** of FIG. **1** and/or the package **200** of FIG. **2**, and may include components that are similar and/or arranged in a similar manner, as described for the package **100** and/or the package **200**.

[0050] The base substrate **302** is similar to the base substrate **102**. The base substrate **302** may include silicon (Si). The base substrate **302** includes a plurality of cavities **320**. The plurality of cavities **320** may be hollow or may be filled with a material (e.g., gas, liquid). The plurality of cavities **320** may be configured as channels for storing fluids. The use of the plurality of cavities **320** may help improve the thermal performance of the package **300**.

[0051] The integrated device **103** is coupled to the base substrate **302**. The integrated device **103** may be coupled to the base substrate **302** through the bonding material **135**. The back side of the integrated device **103** may be coupled to the base substrate **302** through the bonding material **135**. The bonding material **135** may include oxide and/or silicon. The bonding material **135** may be coupled to and touch (i) the base substrate **302** and (ii) the back side of the integrated device **103**. For example the bonding material **135** may be coupled to and touch the die substrate of the integrated device **103**.

[0052] The integrated device **105** is coupled to the base substrate **302**. The integrated device **105** may be coupled to the base substrate **302** through the bonding material **155**. The back side of the integrated device **105** may be coupled to the base substrate **302** through the bonding material **155**. The bonding material **155** may include oxide and/or silicon. The bonding material **155** may be coupled to and touch (i) the base substrate **302** and (ii) the back side of the integrated device **105**. For example the bonding material **155** may be coupled to and touch the die substrate of the integrated device **105**. It is noted that in some implementations, the base substrate **302**, the bonding material **135** and/or the bonding material **155** may be optional.

[0053] The fill material **106** is located between the base substrate **302** and the metallization portion **104**. The fill material **106** may touch the base substrate **302** and the metallization portion **104**. The fill material **106** may at least partially encapsulate the integrated device **103** and the integrated device **105**. The fill material **106** may at least laterally surround the integrated device **103**. The fill material **106** may at least laterally surround the integrated device **105**. The fill material **106** may touch a side surface of the integrated device **103** and/or a side surface of the integrated device **105**. The fill material **106** include inorganic material. The fill material **106** may include silicon oxide.

The fill material **106** may include an encapsulation layer, such as a mold, a resin and/or an epoxy. The fill material **106** may be a means for encapsulation. The fill material **106** may be provided by using a compression and transfer molding process, a sheet molding process, or a liquid molding process.

#### Exemplary Integrated Device

[0054] FIG. **4** illustrates a cross sectional profile view of an integrated device **400** that includes a die substrate. The integrated device **400** may represent the integrated device **103** and/or the integrated device **105**. The integrated device **400** includes a die substrate portion **402** and a die interconnection portion **404**. The die substrate portion **402** includes a die substrate **420**, an active region **422** and a plurality of through substrate vias **421**. The active region **422** may include a plurality of logic cells, a plurality of transistors, and/or a plurality of filters. Different implementations may use different types of transistors, such as a field effect transistor (FET), planar FET, finFET, and a gate all around FET. In some implementations, a front end of line (FEOL) process may be used to fabricate the active region **422** of the die substrate **420**.

[0055] The die substrate **420** may include silicon (Si). The die substrate **420** may comprise a bulk silicon. The bulk silicon may include a monolith silicon. The plurality of through substrate vias **421** may extend through the die substrate **420**. Different implementations may have different thicknesses for the die substrate **420**. In some implementations, the integrated device **400** may include a back side metallization portion that is coupled to the die substrate **420**. The back side metallization portion may include a plurality of back side metallization interconnects that are coupled to the plurality of through substrate vias **421**.

[0056] The die interconnection portion **404** includes at least one dielectric layer **440** and a plurality of die interconnects **442**. The die interconnection portion **404** is coupled to the die substrate portion **402**. The plurality of die interconnects **442** is coupled to the active region **422** of the die substrate portion **402**. The die interconnection portion **404** may also include a plurality of pad interconnects **401** and a passivation layer **406**. In some implementations, a back end of line (BEOL) process may be used to fabricate the die interconnection portion **404**. In some implementations, the process shown in FIGS. **10A-10B** may be used to fabricate the die interconnection portion **404**, including forming the at least one dielectric layer **440** and the plurality of die interconnects **442**. A plurality of metallization interconnects **423** may be coupled to the plurality of through substrate vias **421**. The plurality of metallization interconnects **423** may be part of a back side metallization portion that is formed and coupled to back side of the die substrate **420**.

[0057] In some implementations, an electrical path to and/or from an active region **422** may include at least one die interconnect from the plurality of die interconnects **442**, at least one through substrate via from the plurality of through substrate vias **421**. In some implementations, an electrical path to and/or from an active region **422** may include at least one die interconnect from the plurality of die interconnects **442**, at least one pad interconnect from the plurality of pad interconnects **401**. The integrated device **400** includes a front side and a back side. The front side of the integrated device **400** may be the side that includes the plurality of pad interconnects **401**. The back side of the integrated device **400** may be the side that includes the die substrate **420**, the through substrate vias **421** and/or the plurality of metallization interconnects **423**.

[0058] An integrated device (e.g., **103**, **105**) may include a die (e.g., semiconductor bare die). The integrated device may include a power management integrated circuit (PMIC). The integrated device may include an application processor. The integrated device may include a modem. The integrated device may include a radio frequency (RF) device, a passive device, a filter, a capacitor, an inductor, an antenna, a transmitter, a receiver, a gallium arsenide (GaAs) based integrated device, a surface acoustic wave (SAW) filter, a bulk acoustic wave (BAW) filter, a light emitting diode (LED) integrated device, a silicon (Si) based integrated device, a silicon carbide (SiC) based integrated device, a memory, power management processor, and/or combinations thereof. An integrated device may include at least one electronic circuit (e.g., first electronic circuit, second



electronic circuit, etc . . . ). An integrated device may include an input/output (I/O) hub. An integrated device may include transistors. An integrated device may be an example of an electrical component and/or electrical device.

[0059] In some implementations, an integrated device may be a chiplet. A chiplet may be fabricated using a process that provides better yields compared to other processes used to fabricate other types of integrated devices, which can lower the overall cost of fabricating a chiplet. Different chiplets may have different sizes and/or shapes. Different chiplets may be configured to provide different functions. Different chiplets may have different interconnect densities (e.g., interconnects with different width and/or spacing). In some implementations, several chiplets may be used to perform the functionalities of one or more chips (e.g., one more integrated devices). As mentioned above, using several chiplets that perform several functions may reduce the overall cost of a package relative to using a single chip to perform all of the functions of a package. In some implementations, one or more of the chiplets and/or one of more of integrated devices (e.g., **103**) described in the disclosure may be fabricated using the same technology node or two or more different technology nodes. For example, an integrated device may be fabricated using a first technology node, and a chiplet may be fabricated using a second technology node that is not as advanced as the first technology node. In such an example, the integrated device may include components (e.g., interconnects, transistors) that have a first minimum size, and the chiplet may include components (e.g., interconnects, transistors) that have a second minimum size, where the second minimum size is greater than the first minimum size. In some implementations, a first integrated device and a second integrated device of a package, may be fabricated using the same technology node or different technology nodes. In some implementations, a chiplet and another chiplet of a package, may be fabricated using the same technology node or different technology nodes.

[0060] A technology node may refer to a specific fabrication process and/or technology that is used to fabricate an integrated device and/or a chiplet. A technology node may specify the smallest possible size (e.g., minimum size) that can be fabricated (e.g., size of a transistor, width of trace, gap with between two transistors). Different technology nodes may have different yield loss. Different technology nodes may have different costs. Technology nodes that produce components (e.g., trace, transistors) with fine details are more expensive and may have higher yield loss, than a technology node that produces components (e.g., trace, transistors) with details that are less fine. Thus, more advanced technology nodes may be more expensive and may have higher yield loss, than less advanced technology nodes. When all of the functions of a package are implemented in single integrated devices, the same technology node is used to fabricate the entire integrated device, even if some of the functions of the integrated devices do not need to be fabricated using that particular technology node. Thus, the integrated device is locked into one technology node. To optimize the cost of a package, some of the functions can be implemented in different integrated devices and/or chiplets, where different integrated devices and/or chiplets may be fabricated using different technology nodes to reduce overall costs. For example, functions that require the use of the most advanced technology node may be implemented in an integrated device, and functions that can be implemented using a less advanced technology node can be implemented in another integrated device and/or one or more chiplets. One example, would be an integrated device, fabricated using a first technology node (e.g., most advanced technology node), that is configured to provide compute applications, and at least one chiplet, that is fabricated using a second technology node, that is configured to provide other functionalities, where the second technology node is not as costly as the first technology node, and where the second technology node fabricates components with minimum sizes that are greater than the minimum sizes of components fabricated using the first technology node. Examples of compute applications may include high performance computing and/or high performance processing, which may be achieved by fabricating and packing in as many transistors as possible in an integrated device, which is why an integrated device that is

configured for compute applications may be fabricated using the most advanced technology node available, while other chiplets may be fabricated using less advanced technology nodes, since those chiplets may not require as many transistors to be fabricated in the chiplets. Thus, the combination of using different technology nodes (which may have different associated yield loss) for different integrated devices and/or chiplets, can reduce the overall cost of a package, compared to using a single integrated device to perform all the functions of the package.

[0061] Another advantage of splitting the functions into several integrated devices and/or chiplets, is that it allows improvements in the performance of the package without having to redesign every single integrated device and/or chiplet. For example, if a configuration of a package uses a first integrated device and a first chiplet, it may be possible to improve the performance of the package by changing the design of the first integrated device, while keeping the design of the first chiplet the same. Thus, the first chiplet could be reused with the improved and/or different configured first integrated device. This saves cost by not having to redesign the first chiplet, when packages with improved integrated devices are fabricated.

[0062] The package (e.g., **100, 200, 300**) may be implemented in a radio frequency (RF) package. The RF package may be a radio frequency front end (RFFE) package. A package (e.g., **100, 200, 300**) may be configured to provide Wireless Fidelity (WiFi) communication and/or cellular communication (e.g., 2G, 3G, 4G, 5G). The packages (e.g., **100, 200, 300**) may be configured to support Global System for Mobile (GSM) Communications, Universal Mobile Telecommunications System (UMTS), and/or Long-Term Evolution (LTE). The packages (e.g., **100, 200, 300**) may be configured to transmit and receive signals having different frequencies and/or communication protocols.

Exemplary Sequence for Fabricating a Package Comprising a Base Substrate, Integrated Devices and a Metallization Portion

[0063] In some implementations, fabricating a package includes several processes. FIGS. 5A-5C illustrate an exemplary sequence for providing or fabricating a package. In some implementations, the sequence of FIGS. 5A-5C may be used to provide or fabricate the package **300**.

[0064] It should be noted that the sequence of FIGS. 5A-5C may combine one or more stages in order to simplify and/or clarify the sequence for providing or fabricating a package. In some implementations, the order of the processes may be changed or modified. In some implementations, one or more of processes may be replaced or substituted without departing from the scope of the disclosure. [0065] Stage **1**, as shown in FIG. 5A, illustrates a state after a base substrate **302** is provided. The base substrate **302** may include silicon (Si). The base substrate **302** may include a plurality of cavities **320**. The plurality of cavities **320** may be configured as channels for storing fluid and/or for a fluid to pass through. The plurality of cavities **320** may be hollow or filled with a gas material, a solid material and/or a liquid material. [0066] Stage **2** illustrates a state after the integrated device **103** and the integrated device **105** are coupled to the base substrate **302**. The back side of the integrated device **103** may be coupled to the base substrate **302** through a bonding material **135**. The bonding material **135** may be located between the base substrate **302** and the integrated device **103**. The bonding material **135** may be a first bonding material. The back side of the integrated device **105** may be coupled to the base substrate **302** through a bonding material **155**. The bonding material **155** may be located between the base substrate **302** and the integrated device **105**. The bonding material **155** may be a second bonding material. The bonding material **135** and the bonding material **155** may include the same bonding material. The bonding material **135** and the bonding material **155** may be considered part of the same bonding material. The bonding material **135** and the bonding material **155** may be considered as separate bonding materials. The bonding material **135** and/or the bonding material **155** may include an inorganic material (e.g., silicon oxide, silicon oxide layer). [0067] Stage **3** illustrates a state after a fill material **106** is provided and formed. The fill material **106** may be coupled to and touch the base substrate **302**, the integrated device **103** and the integrated device **105**. The fill material **106** includes inorganic

material. The fill material **106** may include silicon oxide (e.g., silicon oxide layer). In some implementations, the fill material **106** may include the same material (e.g., same inorganic material) as the bonding material **135** and/or the bonding material **155**. In some implementations, the fill material **106** may include a different material (e.g., a different inorganic material) as the bonding material **135** and/or the bonding material **155**.

[0068] In some implementations, the fill material **106** may include an organic material. In some implementations, the fill material **106** may include an encapsulation layer, such as a mold, a resin and/or an epoxy. The fill material **106** may be a means for encapsulation. The fill material **106** may be provided by using a compression and transfer molding process, a sheet molding process, or a liquid molding process. [0069] Stage **4**, as shown in FIG. **5B**, illustrates a state after portions of the fill material **106** are removed. The fill material **106** may be grinded and/or polished. In some implementations, portions of the integrated device **103** and/or portions of the integrated device **105** may also be removed. For example, portions of the front side of the integrated device **103** and/or portions of the front side of the integrated device **105** may be removed. A planarization process may be used to remove portions of the fill material **106**. [0070] Stage **5** illustrates a state after a passive device **247** and a passive device **248** are formed. The passive device **247** may be formed and coupled to the integrated device **103**. For example, the passive device **247** may be formed and coupled to at least one pad from the plurality of pads **130** of the integrated device **103**. A plating process, a deposition process, a lamination process, a lithography process, strip process, an etching process, an exposure process and/or a development process may be used to form the passive device **247**. The passive device **247** may include a metal-insulator-metal (MIM) capacitor. The passive device **248** may be formed and coupled to the integrated device **105**. For example, the passive device **248** may be formed and coupled to at least one pad from the plurality of pads **150** of the integrated device **105**. A plating process, a deposition process, a lamination process, a lithography process, strip process, an etching process, an exposure process and/or a development process may be used to form the passive device **248**. The passive device **248** may include a metal-insulator-metal (MIM) capacitor. [0071] Stage **6** illustrates a state after a metallization portion **104** is formed and coupled to the integrated device **103**, the integrated device **105** and the fill material **106**. The metallization portion **104** may include at least one dielectric layer **140** and a plurality of metallization interconnects **142**. The metallization portion **104** may be formed such that the passive device **247** and/or the passive device **248** are located at least partially in the metallization portion **104**. In some implementations, the metallization portion **104** may include at least one interconnect that is configured as an inductor, such as the metallization interconnect **245** and/or the metallization interconnect **246**. The metallization portion **104** may include a redistribution portion that includes a plurality of redistribution interconnects. FIGS. **7A-7C** illustrate an example of a process for fabricating a metallization portion and/or redistribution portion. FIGS. **10A-10B** illustrate another example of a process for fabricating a metallization portion, where the metallization portion includes damascene interconnects. [0072] Stage **7**, as shown in FIG. **5C**, illustrates a state after the plurality of pillar interconnects **107** are formed and coupled to the metallization portion **104**. The plurality of pillar interconnects **107** are coupled to the plurality of metallization interconnects **142**. A lithography process, a plating process, a strip process and/or an etching process may be used to form the plurality of pillar interconnects **107**. [0073] Stage **8** illustrates a state after a plurality of solder interconnects **109** are coupled to the plurality of pillar interconnects **107**. A solder reflow process may be used to form and couple the plurality of solder interconnects **109** to the plurality of pillar interconnects **107**. Stage **8** may illustrate the package **300** of FIG. **3**.

Exemplary Flow Diagram of a Method for Fabricating a Package Comprising a Base Substrate, Integrated Devices and a Metallization Portion

[0074] In some implementations, fabricating a package includes several processes. FIG. **6** illustrates an exemplary flow diagram of a method **600** for providing or fabricating a package. In some implementations, the method **600** of FIG. **6** may be used to provide or fabricate the package

**300** described in the disclosure. However, the method **600** may be used to provide or fabricate any of the packages (e.g., **100**, **200**) described in the disclosure.

[0075] It should be noted that the method **600** of FIG. **6** may combine one or more processes in order to simplify and/or clarify the method for providing or fabricating a package. In some implementations, the order of the processes may be changed or modified.

[0076] The method provides (at **605**) a base substrate. Stage **1** of FIG. **5A**, illustrates and describes an example of a state after a base substrate **302** is provided. The base substrate **302** may include silicon (Si). The base substrate **302** may include a plurality of cavities **320**. The plurality of cavities **320** may be configured as channels for storing fluid. The plurality of cavities **320** may be hollow or filled with a solid material and/or liquid material.

[0077] The method couples (a **610**) integrated devices to the base substrate through a bonding material. Stage **2** of FIG. **5A**, illustrates and describes an example of a state after the integrated device **103** and the integrated device **105** are coupled to the base substrate **302**. The back side of the integrated device **103** may be coupled to the base substrate **302** through a bonding material **135**. The bonding material **135** may be located between the base substrate **302** and the integrated device **103**. The bonding material **135** may be a first bonding material. The back side of the integrated device **105** may be coupled to the base substrate **302** through a bonding material **155**. The bonding material **155** may be located between the base substrate **302** and the integrated device **105**. The bonding material **155** may be a second bonding material. The bonding material **135** and the bonding material **155** may include the same bonding material. The bonding material **135** and/or the bonding material **155** may include an inorganic material (e.g., silicon oxide, silicon oxide layer).

[0078] The method provides and forms (at **615**) a fill material. Stage **3** of FIG. **5A**, illustrates and describes an example of a state after a fill material **106** is provided and formed. The fill material **106** may be coupled to and touch the base substrate **302**, the integrated device **103** and the integrated device **105**. The fill material **106** includes inorganic material. The fill material **106** may include silicon oxide (e.g., silicon oxide layer). The fill material **106** may include an encapsulation layer, such as a mold, a resin and/or an epoxy. The fill material **106** may be a means for encapsulation. The fill material **106** may be provided by using a compression and transfer molding process, a sheet molding process, or a liquid molding process. An organic material may include polyimide, an underfill and/or a molding compound.

[0079] The method removes (at **620**) a portion of the fill material. Stage **4** of FIG. **5B**, illustrates and describes an example of a state after portions of the fill material **106** are removed. The fill material **106** may be grinded and/or polished. In some implementations, portions of the integrated device **103** and/or portions of the integrated device **105** may also be removed. For example, portions of the front side of the integrated device **103** and/or portions of the front side of the integrated device **105** may be removed.

[0080] The method forms (at **625**) one or more passive devices and/or passive components. Stage **5** of FIG. **5C**, illustrates and describes an example of a state after a passive device **247** and a passive device **248** are formed. The passive device **247** may be formed and coupled to the integrated device **103**. For example, the passive device **247** may be formed and coupled to at least one pad from the plurality of pads **130** of the integrated device **103**. A plating process, a deposition process, a lamination process, a lithography process, strip process, an etching process, an exposure process and/or a development process may be used to form the passive device **247**. The passive device **247** may include a metal-insulator-metal (MIM) capacitor. The passive device **248** may be formed and coupled to the integrated device **105**. For example, the passive device **248** may be formed and coupled to at least one pad from the plurality of pads **150** of the integrated device **105**. A plating process, a deposition process, a lamination process, a lithography process, strip process, an etching process, an exposure process and/or a development process may be used to form the passive device **248**. The passive device **248** may include a metal-insulator-metal (MIM) capacitor.

[0081] The method forms and couples (at **630**) a metallization portion to the fill material and the

integrated devices. Stage 6 of FIG. 5C, illustrates a state after a metallization portion **104** is formed and coupled to the integrated device **103**, the integrated device **105** and the fill material **106**. The metallization portion **104** may include at least one dielectric layer **140** and a plurality of metallization interconnects **142**. The metallization portion **104** may be formed such that the passive device **247** and/or the passive device **248** are located at least partially in the metallization portion **104**. In some implementations, the metallization portion **104** may include at least one interconnect that is configured as an inductor, such as the metallization interconnect **245** and/or the metallization interconnect **246**. The metallization portion **104** may include a redistribution portion that includes a plurality of redistribution interconnects. FIGS. 7A-7C illustrate an example of a process for fabricating a metallization portion and/or redistribution portion. FIGS. 10A-10B illustrate another example of a process for fabricating a metallization portion, where the metallization portion includes damascene interconnects.

[0082] The method forms and couples (at **635**) a plurality of pillar interconnects to the metallization portion. Stage 7 of FIG. 5C, illustrates and describes an example of a state after the plurality of pillar interconnects **107** are formed and coupled to the metallization portion **104**. The plurality of pillar interconnects **107** are coupled to the plurality of metallization interconnects **142**. A lithography process, a plating process, a strip process and/or an etching process may be used to form the plurality of pillar interconnects **107**.

[0083] The method forms and couples (at **640**) a plurality of solder interconnects to the plurality of pillar interconnects. Stage 8 of FIG. 5C, illustrates and describes an example of a state after a plurality of solder interconnects **109** are coupled to the plurality of pillar interconnects **107**. A solder reflow process may be used to form and couple the plurality of solder interconnects **109** to the plurality of pillar interconnects **107**. Stage 8 may illustrate the package **300** of FIG. 3.

#### Exemplary Sequence for Fabricating a Metallization Portion

[0084] In some implementations, fabricating a metallization portion includes several processes. FIGS. 7A-7C illustrate an exemplary sequence for providing or fabricating a metallization portion. In some implementations, the sequence of FIGS. 7A-7C may be used to provide or fabricate the metallization portion **104**. However, the sequence of FIGS. 7A-7C may be used to form any of the metallization portions of the disclosure.

[0085] It should be noted that the sequence of FIGS. 7A-7C may combine one or more stages in order to simplify and/or clarify the sequence for providing or fabricating a metallization portion. In some implementations, the order of the processes may be changed or modified. In some implementations, one or more of processes may be replaced or substituted without departing from the scope of the disclosure. [0086] Stage 1, as shown in FIG. 7A, illustrates a state after a substrate **700** and a silicon oxide layer **710** are provided. The substrate **700** may include a silicon substrate. The silicon oxide layer **710** may include silicon dioxide (SiO<sub>2</sub>). [0087] Stage 2 illustrates a state after a seed layer **720** is formed and coupled to the silicon oxide layer **710**. The seed layer **720** may include copper. The seed layer **720** may be disposed over the silicon oxide layer **710**. [0088] Stage 3 illustrates a state after a photo resist layer **730** is formed and coupled to the seed layer **720**. A deposition process and/or a lamination process may be used to form the photo resist layer **730**. [0089] Stage 4, as shown in FIG. 7B, illustrates a state after a plurality of openings **732** are formed in the photo resist layer **730**. The plurality of openings **732** may expose portions of the seed layer **720**. An exposure process and/or a development process may be used to form the plurality of openings **732**. [0090] Stage 5 illustrates a state after a plurality of interconnects **740** are formed. The plurality of interconnects **740** may be formed in the plurality of openings **732** of the photo resist layer **730**. The plurality of interconnects **740** may be coupled to the seed layer **720**. A plating process may be used to form the plurality of interconnects **740**. [0091] Stage 6 illustrates a state after the photo resist layer **730** is removed. A resist strip process may be used to remove the photo resist layer **730**. [0092] Stage 7 illustrates a state after portions of the seed layer **720** are removed. An etching process may be used to remove portions of the seed layer **720**. In some

implementations, portions of the seed layer **720** that are not covered by the plurality of interconnects **740** may be removed. [0093] Stage **8**, as shown in FIG. **7C**, illustrates a state after a dielectric layer **750** is formed over the silicon oxide layer **710** and/or over the plurality of interconnects **740**. A deposition process and/or a lamination process may be used to form the dielectric layer **750** over the silicon oxide layer **710** and/or the plurality of interconnects **740**. [0094] Stage **9** illustrates a state after a plurality of openings **752** are formed in the dielectric layer **750**. The plurality of openings **752** may expose portions of the plurality of interconnects **740**. An exposure process and/or a development process may be used to form the plurality of openings **752**. [0095] Stage **10** illustrates a state after a plurality of interconnects **760** are formed. The plurality of interconnects **760** may be coupled to the plurality of interconnects **740**. The plurality of interconnects **760** may be located in the plurality of openings **752** of the dielectric layer **750**. The plurality of interconnects **760** may be located over a surface of the dielectric layer **750**. A lithography process, a plating process, a strip process and/or an etching process may be used to form the plurality of interconnects **760**.

[0096] In some implementations, the steps shown in stage **8** through stage **10** may be iteratively performed to form additional dielectric layers and/or additional plurality of interconnects for a metallization portion that includes several metal layers. The plurality of interconnects **740** and/or the plurality of interconnects **760** may include a plurality of metallization interconnects. In some implementations, instead of being formed over the substrate **700** and/or the silicon oxide layer **710**, the plurality of interconnects may be formed over a fill material and/or one or more integrated devices.

[0097] In some implementations, a plurality of metallization interconnects for a metallization portion that are fabricated using the process of FIGS. **7A-7C** may have a minimum width in a range of about 2-10 micrometers. In some implementations, a plurality of metallization interconnects for a metallization portion that are fabricated using the process of FIGS. **7A-7C** may have a minimum space (e.g., minimum spacing) in a range of about 2-10 micrometers.

[0098] Different implementations may have metallization portions with metallization interconnects with different shapes.

[0099] FIG. **8** illustrates a metallization portion **800** that includes at least one dielectric layer **810** and a plurality of metallization interconnects **812**. The plurality of metallization interconnects **812** may include metallization interconnects with different shapes than those shown in FIGS. **1-3** of the disclosure. The metallization portion **800** may include a redistribution portion. The plurality of metallization interconnects **812** may include a plurality of redistribution interconnects. In some implementations, the metallization portion **800** may replace the metallization portion **104**. The plurality of metallization interconnects **812** may replace the plurality of metallization interconnects **142**. The metallization portion **800** may include metallization interconnects that have different shapes than the metallization interconnects of the metallization portion **104**. In some implementations, the metallization portion **800** may be fabricated using a process that is the same and/or similar to the process of FIGS. **7A-7C**.

#### Exemplary Flow Diagram of a Method for Fabricating a Metallization Portion

[0100] In some implementations, fabricating a metallization portion includes several processes. FIG. **9** illustrates an exemplary flow diagram of a method **900** for providing or fabricating a metallization portion. In some implementations, the method **900** of FIG. **9** may be used to provide or fabricate the metallization portion **104** described in the disclosure. However, the method **900** may be used to provide or fabricate any of the metallization portions (e.g., **800**) described in the disclosure.

[0101] It should be noted that the method **900** of FIG. **9** may combine one or more processes in order to simplify and/or clarify the method for providing or fabricating a metallization portion. In some implementations, the order of the processes may be changed or modified.

[0102] The method provides (at **905**) a substrate and a silicon oxide layer. Stage **1** of FIG. **7A**,

illustrates and describes an example of a state after a substrate **700** and a silicon oxide layer **710** is provided. The substrate **700** may include a silicon substrate. The silicon oxide layer **710** may be include silicon dioxide (SiO<sub>2</sub>).

[0103] The method forms (at **910**) a seed layer. Stage **2** of FIG. **7A**, illustrates and describes an example of a state after a seed layer **720** is formed and coupled to the silicon oxide layer **710**. The seed layer **720** may include copper. The seed layer **720** may be disposed over the silicon oxide layer **710**.

[0104] The method forms (at **915**) a photo resist layer with openings. Stage **3** of FIG. **7A**, illustrates and describes an example of a state after a photo resist layer **730** is formed and coupled to the seed layer **720**. A deposition process and/or a lamination process may be used to form the photo resist layer **730**. Stage **4** of FIG. **7B**, illustrates and describes an example of a state after a plurality of openings **732** are formed in the photo resist layer **730**. The plurality of openings **732** may expose portions of the seed layer **720**. An exposure process and/or a development process may be used to form the plurality of openings **732**.

[0105] The method forms (at **920**) a plurality of interconnects. Stage **5** of FIG. **7B**, illustrates and describes an example of a state after a plurality of interconnects **740** are formed. The plurality of interconnects **740** may be formed in the plurality of openings **732** of the photo resist layer **730**. The plurality of interconnects **740** may be coupled to the seed layer **720**. A plating process may be used to form the plurality of interconnects **740**.

[0106] The method removes (at **925**) the photo resist layer and part of the seed layer. Stage **6** of FIG. **7B**, illustrates and describes an example of a state after the photo resist layer **730** is removed. A resist strip process may be used to remove the photo resist layer **730**. Stage **7** of FIG. **7B**, illustrates and describes an example of a state after portions of the seed layer **720** are removing. An etching process may be used to remove portions of the seed layer **720**. In some implementations, portions of the seed layer **720** that are not covered by the plurality of interconnects **740** may be removed.

[0107] The method forms (at **930**) a dielectric layer with openings. Stage **8** of FIG. **7C**, illustrates and describes an example of a state after a dielectric layer **750** is formed over the silicon oxide layer **710** and/or over the plurality of interconnects **740**. A deposition process and/or a lamination process may be used to form the dielectric layer **750** over the silicon oxide layer **710** and/or the plurality of interconnects **740**.

[0108] Stage **9** of FIG. **7C**, illustrates and describes an example of a state after a plurality of openings **752** are formed in the dielectric layer **750**. The plurality of openings **752** may expose portions of the plurality of interconnects **740**. An exposure process and/or a development process may be used to form the plurality of openings **752**.

[0109] The method forms (at **935**) a plurality of interconnects. Stage **10** of FIG. **7C**, illustrates and describes an example of a state after a plurality of interconnects **760** are formed. The plurality of interconnects **760** may be coupled to the plurality of interconnects **740**. The plurality of interconnects **760** may be located in the plurality of openings **752** of the dielectric layer **750**. The plurality of interconnects **760** may be located over a surface of the dielectric layer **750**. A lithography process, a plating process, a strip process and/or an etching process may be used to form the plurality of interconnects **760**.

[0110] In some implementations, forming (at **930**) dielectric layers with openings and forming (at **935**) a plurality of interconnects may be iteratively performed to form additional dielectric layers and/or additional plurality of interconnects for a metallization portion that includes several metal layers. The plurality of interconnects **740** and/or the plurality of interconnects **760** may include a plurality of metallization interconnects. In some implementations, instead of being formed over the substrate **700** and/or the silicon oxide layer **710**, the plurality of interconnects may be formed over a fill material and/or one or more integrated devices.

Exemplary Sequence for Fabricating a Metallization Portion

[0111] In some implementations, fabricating a metallization portion includes several processes. FIGS. **10A-10B** illustrate an exemplary sequence for providing or fabricating a metallization portion. In some implementations, the sequence of FIGS. **10A-10B** may be used to provide or fabricate the metallization portion **104**. The metallization portion **104** may be fabricated using a damascene process, which may provide high density interconnects (e.g., damascene interconnects). [0112] It should be noted that the sequence of FIGS. **10A-10B** may combine one or more stages in order to simplify and/or clarify the sequence for providing or fabricating a metallization portion. In some implementations, the order of the processes may be changed or modified. In some implementations, one or more of processes may be replaced or substituted without departing from the scope of the disclosure. [0113] Stage **1**, as shown in FIG. **10A**, illustrates a state after a dielectric layer **1000**, a liner layer **1002**, an interconnect **1004**, a coating layer **1006**, a dielectric layer **1008**, a coating layer **1010**, a photo resist layer **1020**, and an opening **1012** are provided and formed. The dielectric layer **1000** and/or the dielectric layer **1008** may include SiCOH (e.g., Si, C, O and H). The coating layer **1006** may include silicon carbon nitride (SiCN). The coating layer **1006** may help protect the interconnect **1004** from oxidation. The coating layer **1010** may include silicon oxide (e.g., silicon dioxide). The liner layer **1002** may include a barrier layer (e.g., TaN/Ta). The liner layer **1002** may help prevent metal from the interconnect **1004** from diffusing into the dielectric layer **1000**. Stage **1** illustrates the opening **1012** being formed through the coating layer **1010**, the dielectric layer **1008** and/or the coating layer **1006**. [0114] Stage **2** illustrates a state after (i) the photo resist layer **1020** is removed, and (ii) an anti-reflection coating layer **1030** is formed, and another photo resist layer **1040** is formed. The anti-reflection coating layer **1030** is formed in the opening **1012** and over the coating layer **1010**. The photo resist layer **1040** is formed over portions of the anti-reflection coating layer **1030**. [0115] Stage **3** illustrates a state after portions of the anti-reflection coating layer **1030** are removed. Stage **3** also illustrates a state after an opening **1018** is formed in the dielectric layer **1008**. [0116] Stage **4**, as shown in FIG. **10B**, illustrates a state after a liner layer **1050** and a seed layer **1060** are formed. The liner layer **1050** may include a barrier layer (e.g., TaN/Ta). The seed layer **1060** is formed over the liner layer **1050**. The seed layer **1060** may include copper. A deposition process may be used to form the liner layer **1050**. A deposition process may be used to form the seed layer **1060**. [0117] Stage **5** illustrates a state after an interconnect **1070** is formed. The interconnect **1070** may include copper. The interconnect **1070** may be coupled to the seed layer **1060**. A plating process may be used to form the interconnect **1070**. [0118] Stage **6** illustrates a state after the interconnect **1070** is planarized and a coating layer **1080** is formed over the interconnect **1070** and the dielectric layer **1008**. The coating layer **1080** may include silicon carbon nitride (SiCN). The process of stages **1** through **6** may be iteratively repeated to form additional dielectric layers and/or additional interconnects. [0119] It is noted that the liner layer **1050** may be considered part of the interconnect **1070**. Thus, in some implementations, a liner layer that is touching an interconnect may be considered part of the interconnect. Similarly, a seed layer that is touching an interconnect may be considered part of the interconnect. [0120] It is noted that any of the electrically conductive material may be considered part of an interconnect. Thus, an electrically conductive material that touches the interconnect may be considered part of the interconnect. For example, the seed layer **1060** may be considered part of the interconnect **1070**. The plurality of interconnects that are formed using the above process may have a minimum pitch of about 25 nanometers (nm). In some implementations, interconnects may have thickness in a range of about 20 nanometers-5 micrometers. In some implementations, a dielectric may have a thickness in a range of about 20 nanometers-5 micrometers. For example, a dielectric layer between two adjacent metal layers of the plurality of interconnects may have a thickness in a range of about 20 nanometers-5 micrometers. [0121] Thus, a metallization portion that is fabricated using the above damascene process may have interconnects with a minimum pitch of about 25 nanometers (nm).



#### Exemplary Electronic Devices

[0122] FIG. 11 illustrates various electronic devices that may be integrated with any of the aforementioned device, integrated device, integrated circuit (IC) package, integrated circuit (IC) device, semiconductor device, integrated circuit, die, interposer, package, package-on-package (POP), System in Package (SiP), or System on Chip (SoC). For example, a mobile phone device 1102, a laptop computer device 1104, a fixed location terminal device 1106, a wearable device 1108, or automotive vehicle 1110 may include a device 1100 as described herein. The device 1100 may be, for example, any of the devices and/or integrated circuit (IC) packages described herein. The devices 1102, 1104, 1106 and 1108 and the vehicle 1110 illustrated in FIG. 11 are merely exemplary. Other electronic devices may also feature the device 1100 including, but not limited to, a group of devices (e.g., electronic devices) that includes mobile devices, hand-held personal communication systems (PCS) units, portable data units such as personal digital assistants, global positioning system (GPS) enabled devices, navigation devices, set top boxes, music players, video players, entertainment units, fixed location data units such as meter reading equipment, communications devices, smartphones, tablet computers, computers, wearable devices (e.g., watches, glasses), Internet of things (IoT) devices, servers, routers, electronic devices implemented in automotive vehicles (e.g., autonomous vehicles), or any other device that stores or retrieves data or computer instructions, or any combination thereof.

[0123] One or more of the components, processes, features, and/or functions illustrated in FIGS. 1-4, 5A-5C, 6, 7A-7C, 8-9, 10A-10B and 11 may be rearranged and/or combined into a single component, process, feature or function or embodied in several components, processes, or functions. Additional elements, components, processes, and/or functions may also be added without departing from the disclosure. It should also be noted FIGS. 1-4, 5A-5C, 6, 7A-7C, 8-9, 10A-10B and 11 and its corresponding description in the present disclosure is not limited to dies and/or ICs. In some implementations, FIGS. 1-4, 5A-5C, 6, 7A-7C, 8-9, 10A-10B and 11 and its corresponding description may be used to manufacture, create, provide, and/or produce devices and/or integrated devices. In some implementations, a device may include a die, an integrated device, an integrated passive device (IPD), a die package, an integrated circuit (IC) device, a device package, an integrated circuit (IC) package, a wafer, a semiconductor device, a package-on-package (POP) device, a heat dissipating device and/or an interposer.

[0124] It is noted that the figures in the disclosure may represent actual representations and/or conceptual representations of various parts, components, objects, devices, packages, integrated devices, integrated circuits, and/or transistors. In some instances, the figures may not be to scale. In some instances, for purpose of clarity, not all components and/or parts may be shown. In some instances, the position, the location, the sizes, and/or the shapes of various parts and/or components in the figures may be exemplary. In some implementations, various components and/or parts in the figures may be optional.

[0125] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation or aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects of the disclosure. Likewise, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation. The term “coupled” is used herein to refer to the direct or indirect coupling (e.g., mechanical coupling) between two objects. For example, if object A physically touches object B, and object B touches object C, then objects A and C may still be considered coupled to one another-even if they do not directly physically touch each other. An object A, that is coupled to an object B, may be coupled to at least part of object B. The term “electrically coupled” may mean that two objects are directly or indirectly coupled together such that an electrical current (e.g., signal, power, ground) may travel between the two objects. Two objects that are electrically coupled may or may not have an electrical current traveling between the two objects. The use of the terms “first”, “second”, “third” and “fourth” (and/or anything above fourth) is arbitrary. Any of the

components described may be the first component, the second component, the third component or the fourth component. For example, a component that is referred to a second component, may be the first component, the second component, the third component or the fourth component. The terms “encapsulate”, “encapsulating” and/or any derivation means that the object may partially encapsulate or completely encapsulate another object. The terms “top” and “bottom” are arbitrary. A component that is located on top may be located over a component that is located on a bottom. A top component may be considered a bottom component, and vice versa. As described in the disclosure, a first component that is located “over” a second component may mean that the first component is located above or below the second component, depending on how a bottom or top is arbitrarily defined. In another example, a first component may be located over (e.g., above) a first surface of the second component, and a third component may be located over (e.g., below) a second surface of the second component, where the second surface is opposite to the first surface. It is further noted that the term “over” as used in the present application in the context of one component located over another component, may be used to mean a component that is on another component and/or in another component (e.g., on a surface of a component or embedded in a component). Thus, for example, a first component that is over the second component may mean that (1) the first component is over the second component, but not directly touching the second component, (2) the first component is on (e.g., on a surface of) the second component, and/or (3) the first component is in (e.g., embedded in) the second component. A first component that is located “in” a second component may be partially located in the second component or completely located in the second component. A value that is about X-XX, may mean a value that is between X and XX, inclusive of X and XX. The value(s) between X and XX may be discrete or continuous. The term “about ‘value X’”, or “approximately value X”, as used in the disclosure means within 10 percent of the ‘value X’. For example, a value of about 1 or approximately 1, would mean a value in a range of 0.9-1.1.

[0126] In some implementations, an interconnect is an element or component of a device or package that allows or facilitates an electrical connection between two points, elements and/or components. In some implementations, an interconnect may include a trace (e.g., trace interconnect), a via (e.g., via interconnect), a pad (e.g., pad interconnect), a pillar, a metallization layer, a redistribution layer, and/or an under bump metallization (UBM) layer/interconnect. In some implementations, an interconnect may include an electrically conductive material that may be configured to provide an electrical path for a signal (e.g., a data signal), ground and/or power. An interconnect may include more than one element or component. An interconnect may be defined by one or more interconnects. An interconnect may include one or more metal layers. An interconnect may be part of a circuit. Different implementations may use different processes and/or sequences for forming the interconnects. In some implementations, a chemical vapor deposition (CVD) process, a physical vapor deposition (PVD) process, a sputtering process, a spray coating, and/or a plating process may be used to form the interconnects.

[0127] Also, it is noted that various disclosures contained herein may be described as a process that is depicted as a flowchart, a flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed.

[0128] In the following, further examples are described to facilitate the understanding of the invention. [0129] Aspect 1: A package comprising a base substrate; a first integrated device coupled to the base substrate through a bonding material; a second integrated device coupled to the base substrate through the bonding material; a fill material coupled to the base substrate, the first integrated device and the second integrated device; and a metallization portion coupled to the first integrated device and the second integrated device. [0130] Aspect 2: The package of aspect 1, wherein the bonding material comprises a first bonding material and a second bonding material,

wherein the first integrated device comprises a first back side that is coupled to and touching the first bonding material, and wherein the second integrated device comprises a second back side that is coupled to and touching the second bonding material. [0131] Aspect 3: The package of aspects 1 through 2, wherein the fill material is coupled to and touching the first integrated device, the second integrated device and the base substrate. [0132] Aspect 4: The package of aspects 1 through 3, wherein the metallization portion comprises at least one passive device. [0133] Aspect 5: The package of aspect 4, wherein the at least one passive device comprises a capacitor and/or an inductor. [0134] Aspect 6: The package of aspects 1 through 5, further comprising a plurality of pillar interconnects coupled to the metallization portion. [0135] Aspect 7: The package of aspects 1 through 6, wherein the metallization portion comprises at least one dielectric layer; and a plurality of metallization interconnects. [0136] Aspect 8: The package of aspect 7, wherein the plurality of metallization interconnects comprise a plurality of damascene interconnects. [0137] Aspect 9: The package of aspects 7 through 8, wherein the plurality of metallization interconnects comprises interconnects with width and spacing that are less than 1 micron. [0138] Aspect 10: The package of aspects 1 through 9, wherein the base substrate includes at least one cavity, wherein the fill material comprises a first inorganic material, and wherein the bonding material comprises a second inorganic material. [0139] Aspect 11: The package of aspects 1 through 10, wherein the base substrate comprises silicon, wherein the fill material comprises silicon oxide, and wherein the bonding material comprises oxide or silicon. [0140] Aspect 12: A method for fabricating a package. The method provides a base substrate. The method couples a first integrated device to the base substrate through a bonding material. The method couples a second integrated device to the base substrate through the bonding material. The method couples a fill material to the base substrate, the first integrated device and the second integrated device. The method couples a metallization portion to the first integrated device and the second integrated device. [0141] Aspect 13: The method of aspect 12, wherein the first integrated device comprises a first back side that is coupled to and touching the bonding material, and wherein the second integrated device comprises a second back side that is coupled to and touching the bonding material. [0142] Aspect 14: The method of aspects 12 through 13, wherein the fill material is coupled to and touching the first integrated device, the second integrated device and the base substrate. [0143] Aspect 15: The method of aspects 12 through 14, wherein the metallization portion comprises at least one passive device. [0144] Aspect 16: The method of aspects 12 through 15, wherein the metallization portion comprises at least one dielectric layer; and a plurality of metallization interconnects. [0145] Aspect 17: The method of aspect 16, wherein the plurality of metallization interconnects comprise a plurality of damascene interconnects. [0146] Aspect 18: The method of aspects 16 through 17, wherein the plurality of metallization interconnects comprises interconnects with width and spacing that are less than 1 micron. [0147] Aspect 19: The method of aspects 12 through 18, wherein the base substrate includes at least one cavity. [0148] Aspect 20: The method of aspects 12 through 19, wherein the base substrate comprises silicon, wherein the fill material comprises silicon oxide, and wherein the bonding material comprises oxide or silicon. [0149] Aspect 21: The package of aspects 1 through 11, wherein the package is implemented in a device that is selected from a group consisting of a music player, a video player, an entertainment unit, a navigation device, a communications device, a mobile device, a mobile phone, a smartphone, a personal digital assistant, a fixed location terminal, a tablet computer, a computer, a wearable device, a laptop computer, a server, an internet of things (IoT) device, and a device in an automotive vehicle.

[0150] The various features of the disclosure described herein can be implemented in different systems without departing from the disclosure. It should be noted that the foregoing aspects of the disclosure are merely examples and are not to be construed as limiting the disclosure. The description of the aspects of the present disclosure is intended to be illustrative, and not to limit the scope of the claims. As such, the present teachings can be readily applied to other types of

apparatuses and many alternatives, modifications, and variations will be apparent to those skilled in the art.

## Claims

1. A package comprising: a base substrate; a first integrated device coupled to the base substrate through a bonding material; a second integrated device coupled to the base substrate through the bonding material; a fill material coupled to the base substrate, the first integrated device and the second integrated device; and a metallization portion coupled to the first integrated device and the second integrated device.
2. The package of claim 1, wherein the bonding material comprises a first bonding material and a second bonding material, wherein the first integrated device comprises a first back side that is coupled to and touching the first bonding material, and wherein the second integrated device comprises a second back side that is coupled to and touching the second bonding material.
3. The package of claim 1, wherein the fill material is coupled to and touching the first integrated device, the second integrated device and the base substrate.
4. The package of claim 1, wherein the metallization portion comprises at least one passive device.
5. The package of claim 4, wherein the at least one passive device comprises a capacitor and/or an inductor.
6. The package of claim 1, further comprising a plurality of pillar interconnects coupled to the metallization portion.
7. The package of claim 1, wherein the metallization portion comprises: at least one dielectric layer; and a plurality of metallization interconnects.
8. The package of claim 7, wherein the plurality of metallization interconnects comprise a plurality of damascene interconnects.
9. The package of claim 7, wherein the plurality of metallization interconnects comprises interconnects with width and/or spacing that are less than 1 micron.
10. The package of claim 1, wherein the base substrate includes at least one cavity, wherein the fill material comprises a first inorganic material, and wherein the bonding material comprises a second inorganic material.
11. The package of claim 1, wherein the base substrate comprises silicon, wherein the fill material comprises silicon oxide, and wherein the bonding material comprises oxide or silicon.
12. A method for fabricating a package, comprising: providing a base substrate; coupling a first integrated device to the base substrate through a bonding material; coupling a second integrated device to the base substrate through the bonding material; coupling a fill material to the base substrate, the first integrated device and the second integrated device; and coupling a metallization portion to the first integrated device and the second integrated device.
13. The method of claim 12, wherein the first integrated device comprises a first back side that is coupled to and touching the bonding material, and wherein the second integrated device comprises a second back side that is coupled to and touching the bonding material.
14. The method of claim 12, wherein the fill material is coupled to and touching the first integrated device, the second integrated device and the base substrate.
15. The method of claim 12, wherein the metallization portion comprises at least one passive device.
16. The method of claim 12, wherein the metallization portion comprises: at least one dielectric layer; and a plurality of metallization interconnects.
17. The method of claim 16, wherein the plurality of metallization interconnects comprise a plurality of damascene interconnects.
18. The method of claim 16, wherein the plurality of metallization interconnects comprises interconnects with width and spacing that are less than 1 micron.

- 19.** The method of claim 12, wherein the base substrate includes at least one cavity.
- 20.** The method of claim 12, wherein the base substrate comprises silicon, wherein the fill material comprises silicon oxide, and wherein the bonding material comprises oxide or silicon.
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